OPTIMIZATION - 2

Class - 2

- Agenda →
  - Recap
  - Limits & CONTINUITY ⇒ Last Class
  - DERIVATIVES
  - Differentiability
  - Common Derivatives
  - Rules → Product Rule, Quotient Rule, Chain Rule
  - Maxima & Minima ⇒ optimization

Recap ⇒

LA module ⇒

= optimization for Gain function

→ max. gain function

Given → D, $\vec{w}$, w_0

line

(+) divide — line

(+) $\underset{w, w_0}{\text{argmax}} \; q(D, \vec{w}, w_0)$

$$w^*, w_0^* = \text{argmax}\, q(D, w, w_0)$$

(*) Objective → maximise

(*) ML problems ⇒ trip this

Loss ⇒ (*) (*)

↓ Error

(*) Loss Min. Objective?

gain → max

→ Back,

$$w^*, w_0 = \underset{}{\text{argmin}}\; \text{Loss}(D, w, w_0)$$

Loss objective

① $f(x) = (x-5)^2$ ⇒

gain → max
loss → min.

② min value?

$f(5) = (5-5)^2 = 0$ ⇒ ④ $x = 5$

$x = 5$ ⑧ ⇒

$x = -5$

$(-5-5)^2 = (-10)^2 = 100$

$-1, -2 \Rightarrow$ (70)

other terms = $x = 8, 2, 1$

$f(x) > 0$



able to visualize

(1)

③ 100 variables ⇒ we cant visualize

(*) Tool ⇒ minimum ⇒ CALCULUS

↖ Optimisation

(*) gain func^n ⇒ ↑ distances

maximising

(*) ↓

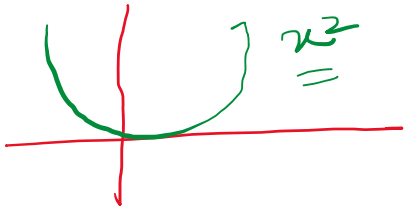
Loss ⇒ ↓ Loss ⇒ minimising

(*) Limits ⇒

functions ↗ CONTINUOUS

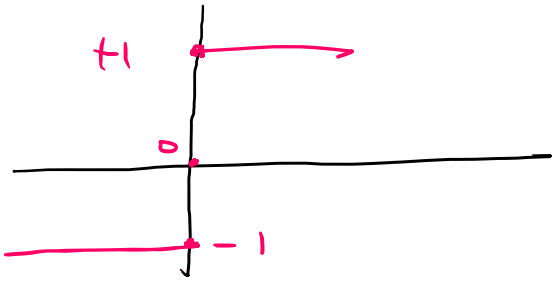
↘ Non CONTINUOUS

CONTINUOUS func^n ⇒ can be drawn without lifting pen from paper



$x+3$, cosine, parabola, $|x|$

Non continuous fun^{n} ⇒



(*) $y = f(x) = \begin{cases} +1 ; x > 0 \\ -1 ; x < 0 \\ 0 ; x = 0 \end{cases}$

$y = \frac{1}{x}$, $\tan x$

Limits ⇒ approaching a number

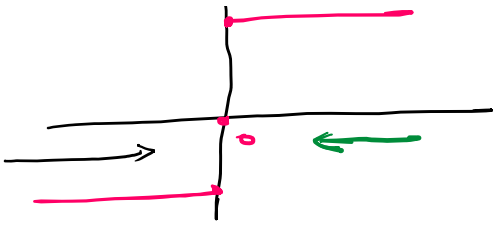
(1) RHL ⇒ no from Right side (+ve dir.)

(2) LHL ⇒ no from left side (-ve dir.)

$y = f(x) = \begin{cases} +1 ; x > 0 \\ -1 ; x < 0 \\ 0 ; x = 0 \end{cases}$

RHL

(*) $\lim_{x \to 0^+} f(x) = +1$



from the side of

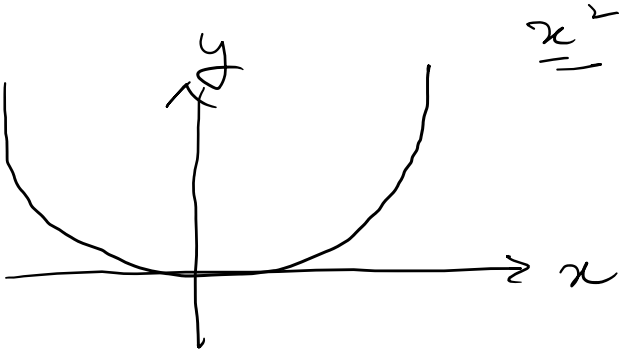
(*) $x = 0.1, 0.01, \overline{0.001}$ ...
But not exactly 0

$-0.001, -0.01, -0.1$

LHL $\Rightarrow$ $\lim_{x \to 0^-} f(x) = -1$

RHL $\neq$ LHL

* Diff. func^n $\Rightarrow$



x^2

$x = 0$

$f(x) = x^2$

RHL

$\lim_{x \to 0^+} f(x) = 0$

LHL

$\lim_{x \to 0^-} f(x) = 0$

$\Rightarrow$

$(3)^2 \quad (-3)^2$

9

$\lim_{x \to 3^+} f(x) = 9$

$\lim_{x \to 3^-} f(x) = 9$

$1+0.1 \quad (-0.1)^2$

RHL = LHL

$(+0.1)^2$ $(-0.1)^2$

x^2

(*) Difference

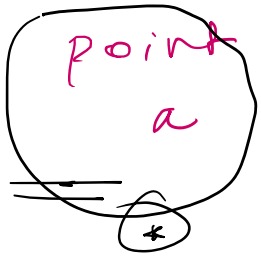
Why in 2nd case Both LHL = RHL & not in 1st case?

(*) LHL = RHL ⇒ func^n more CONTINUOUS

Limits ⇒ func^n continuous or not

(*)

func^n is continuous at a point if ⇒



LHL	RHL	$f(x)_{at\ a}$
$\lim_{x \to a^-} f(x)$	$\lim_{x \to a^+} f(x)$	$f(a)$

(*) Continuity ⟹ Differentiability

(*) Differentiability ⟹ Continuity

0

RHL LHL

(*) $f(x) = \begin{cases} 1 & ; x \geq 2 \\ 0 & ; x < 2 \end{cases}$

3

$(+3)^2 = (-3)^2$

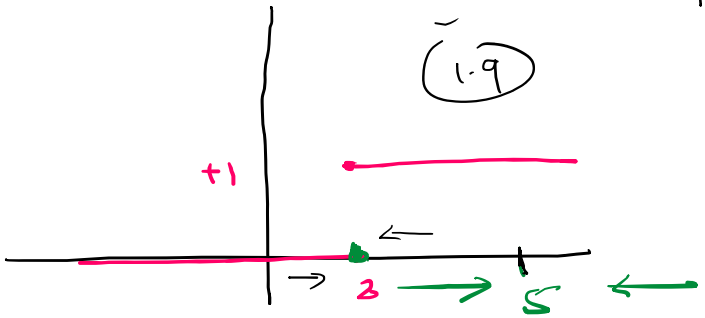
$a = 2$

1.9

$1.9 < 2 \Rightarrow 0$

$2.1 > 2 \Rightarrow 1$

CONTINUOUS OR NOT ?



1

1 → 2 → 3 ←

4.9 → 1 ①

$f(3) = 1$

NO

LHL & RHL

Point $a = 3$

LHL

$\lim_{x \to 3^-} f(x) = 1$

RHL

$\lim_{x \to 3^+} f(x) = 1$

$f(3) = 1$

LHL = RHL = $f(3)$

✓

We proved fun's as CONTINUOUS

fun^n is continuous at $x = 3$

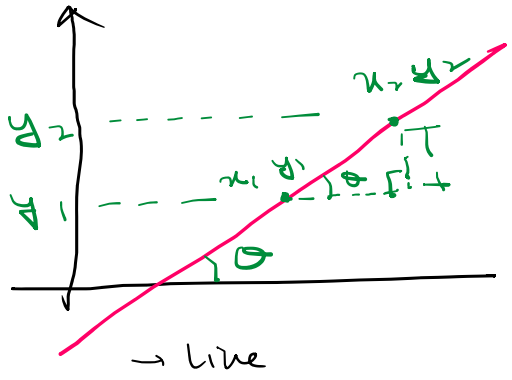
→ Not continuous at $x = 2$



continuous functions are continuous everywhere

⑦ DERIVATIVES

Differentiation ⇒ geometric intuition



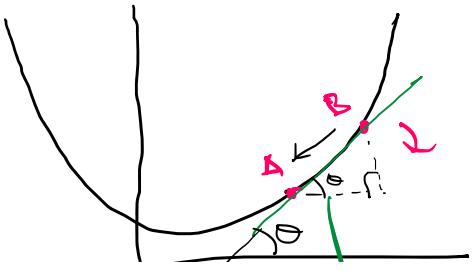
slope of line $= \frac{y_2 - y_1}{x_2 - x_1}$

$\tan\theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1}$

2 points ⇒ slope of line

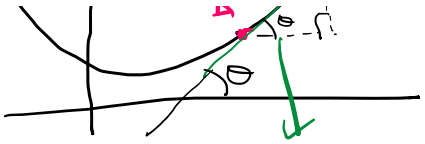
x^2

$y = f(x)$



* $A \rightarrow (x_1, y_1) \rightarrow (x_1, f(x_1))$

$B \rightarrow x_2 \; y_2 \rightarrow (x_2, f(x_2))$



B → x_2 y_2 → $(x_2, f(x_2))$

(*) Secant

$$\tan\theta = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

(*) avg. Rate of change

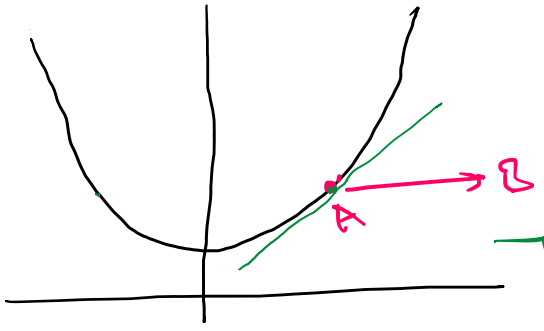
$y_1 \to f(x_1)$

$y_2 \to f(x_2)$

$x_2 \to$

B → x_1

B → x_1



infinitely

$x_2 \to x_1$

↳

Secant → tangent

geometrically

(*) Tangent

↳ touching line exactly at 1 point

we can say

$\Delta x \rightarrow$ very small

$x_2 = x_1 + 0.0001 \rightarrow \Delta$

small value

Tangent Relation to Derivatives!

slope of tangent = Derivative of func" at that point

(x_1, y_1)

$x_2 = x_1 + \underbrace{0.0001}_{\Delta x}$

$$m = \frac{f(x_1 + \Delta x) - f(x_1)}{x_1 + \Delta x - x_1}$$

$$m = f(x_1 + \Delta x) - f(x_1)$$

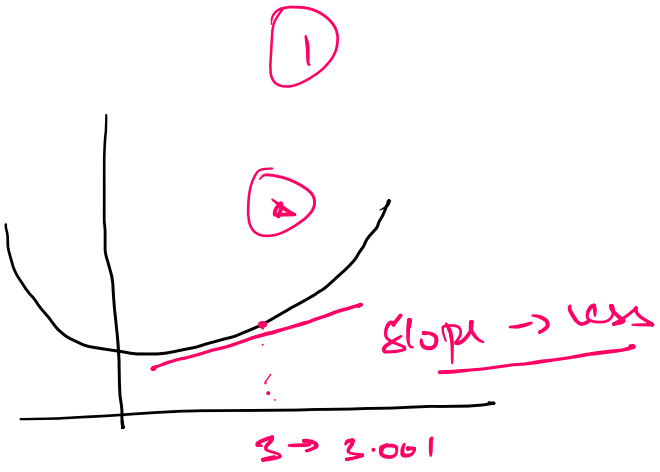
$$m = \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

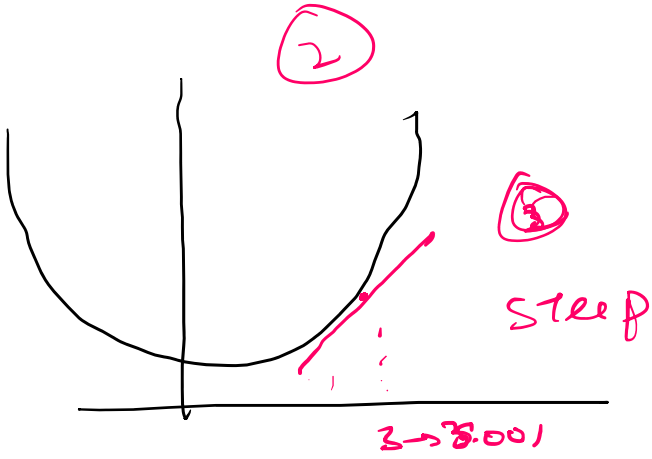
1st principle way

derivative of any function

derivative = slope of tangent

↳ Rate of change w.r.t. x





Change will be higher ↓ quick

2 ⇒ 2

Rate of change = $\frac{\Delta f}{\Delta x}$

$\Delta x \to 0$ → derivative

$\frac{df}{dx}$

→ Derivative of funⁿ at diff. points will be different

given → $f(x)$

$$\frac{df(x)}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

10:25

How to find derivative of funⁿ ⇒

How to find derivative of funⁿ ⇒

$f(x) = x^2 \Rightarrow 2x$ ✓

$= nx^{n-1}$ ✓

$= 2x$

$y = f(x) = x^2$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$\Rightarrow \lim_{\Delta x \to 0} \frac{(x+\Delta x)^2 - x^2}{\Delta x}$$

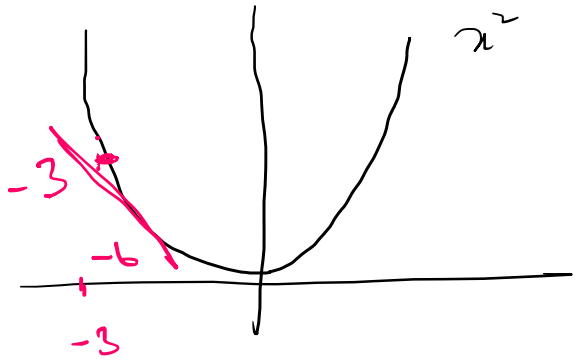
$$\Rightarrow \lim_{\Delta x \to 0} \frac{x^2 + \Delta x^2 + 2x\Delta x - x^2}{\Delta x}$$

$$\lim_{\Delta x \to 0} \frac{\Delta x(\Delta x + 2x)}{\Delta x}$$

$$\lim \quad \Delta x + 2x$$

$\Delta x \to 0$

$= 2x$



$$\left.\frac{d}{dx}(x^2)\right|_{x=2} = 2 \times 2 = 4$$

Steep slope

$$\left.\frac{d}{dx}(x^2)\right|_{x=-3} = 2 \times -3 = -6$$

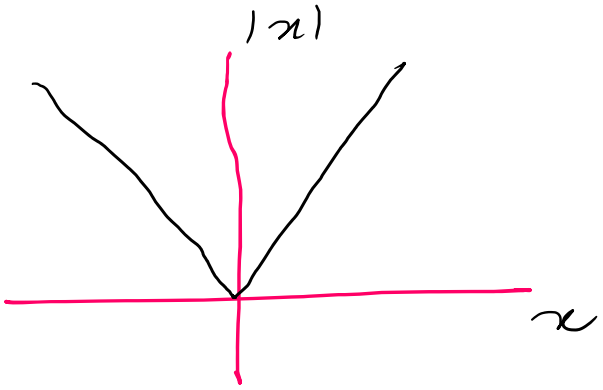
$$\boxed{\frac{d}{dx}(x^n) = n \cdot x^{n-1}}$$

x^2

$$f(x) = |x| = \begin{cases} x\,; & x > 0 \\ -x\,; & x < 0 \\ 0\,; & x = 0 \end{cases}$$

continuous ?

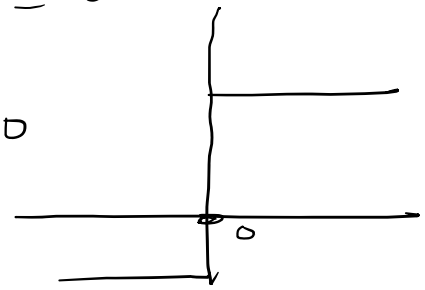
$f(x)$



LHL $\Rightarrow x \to 0^-$ of $|x| = 0$

RHL $\Rightarrow x \to 0^+ = 0$

$f(0) = 0$



$\Rightarrow$ continuous at $x = 0$

Case 1 $\Rightarrow x > 0$

$f(x) = x$

$f'(x) = 1$

Case 2 $\Rightarrow x < 0$

$f(x) = -x$

$f'(x) = -1$

Case 3 $\Rightarrow x = 0$

$\therefore f(x) \neq \lim f(x)$

↓ $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$

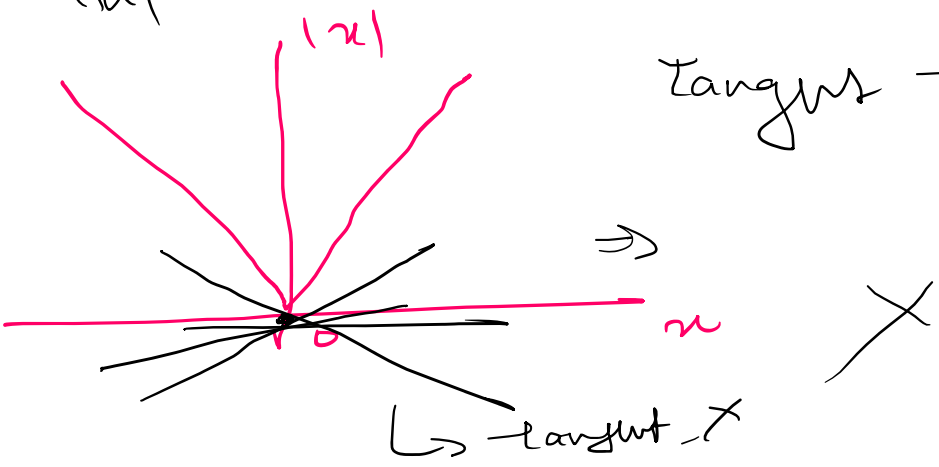
$= -1 \neq +1$

(*) Derivative does not exist at $x = 0$

$|x| \Rightarrow$ X at $x = 0$

X

|x|



Tangent → touches 1 point only

⇒ X

(*) Test for Differentiability

(1) get $f'(x)$

① get $f'(x)$

if $f'(x)$ is discontinuous

↳ fun^{n} is not diff. at the point

Differentiability = Continuous + Smoothness

Commonly used DERIVATIVES →

① $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$ ② $\frac{d}{dx}(\log x) = \frac{1}{x}$

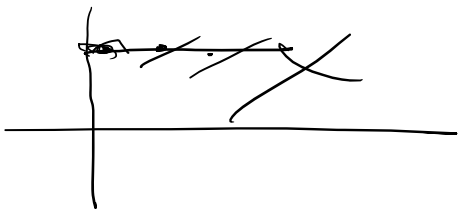
③ $\frac{d}{dx}(e^x) = e^x$ ④ $\frac{d}{dx}(2) = 0$

↓

constant

constants

(5) $\frac{d}{dx}(\sin x) = \cos x$



(6) $\frac{d}{dx}(\cos x) = -\sin x$

(4) Rules of DERIVATIVES

(1) Linearity Rule ⇒ → Linear

$h(x) = f(x) \pm g(x)$

$\Rightarrow h'(x) = f'(x) \pm g'(x)$ ⇒ Answer

Linearity rule

$h(x) = \sin x - x^2$

↓

$h'(x) = \cos x - 2x$

Linearity Rule

(4)

Linearity Rule

(2) Product Rule ⇒ (4)

$h(x) = f(x) \cdot g(x)$

(4) $h'(x) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$

$f(x) = x \cdot \log x$ ($f(x)$ → x, $g(x)$ → $\log x$)

$f'(x) = 1 \cdot \log x + \frac{1}{x} \cdot x$

$= 1 + \log x$

$f(x) = \frac{\log x}{x} \Rightarrow \log x \cdot \frac{1}{x}$ ($f(x)$ → $\log x$, $g(x)$ → $\frac{1}{x}$)

$\frac{1}{x} \to x^{-1}$

$$f'(x) = \frac{1 - \log x}{x^2} \qquad \log x \cdot x^{-1}$$

$$\Rightarrow \frac{1}{x} \times x^{-1} + (-x^{-2}) \cdot \log x$$

$$\Rightarrow \left(\frac{1 - \log x}{x^2} \right)$$

(2) Quotient rule

$$h(x) = \frac{f(x)}{g(x)} \Rightarrow f(x) \cdot \frac{1}{g(x)}$$

(1)

$$h'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

(3) $\Rightarrow$ Basic

(4) Chain Rule ⇒ entire ML about this

$$h(x) = f(g(x))$$

① ② ③ $g(x)$ $f''(x)$

$\times$ $f(x)$ $\Rightarrow$ $\times$

$$\boxed{h'(x) = f'(g(x)) \cdot g'(x)}$$

$$h(x) = \log(x^2)$$

$f = \log$ $\quad$ $g = x^2$

$$\frac{d}{dx^2}(\log x^2) \cdot \frac{dx^2}{dx}$$

$$\frac{d(\log x^2)}{dx} \cdot \frac{dx}{dx}$$

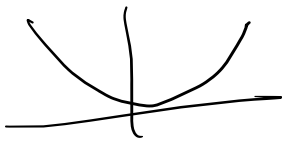
$$\frac{1}{x^2} \cdot 2x = \frac{2}{x} \quad (*)$$

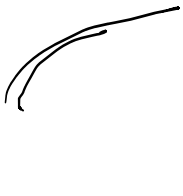
Derivatives → code

1st func.

$\log(x^2)$ $\quad$ $(\log x)^2$

$\Rightarrow 2 \log x$

$x \rightarrow$ 

$\log$ 

$\Rightarrow$

$f(x) = \sin(e^x)$

f : $\sin$ $\quad$ g : e^x

$f(g(x))$

$$= \left(\frac{d(\sin(e^x))}{de^x}\right) \cdot \frac{d(e^x)}{dx}$$

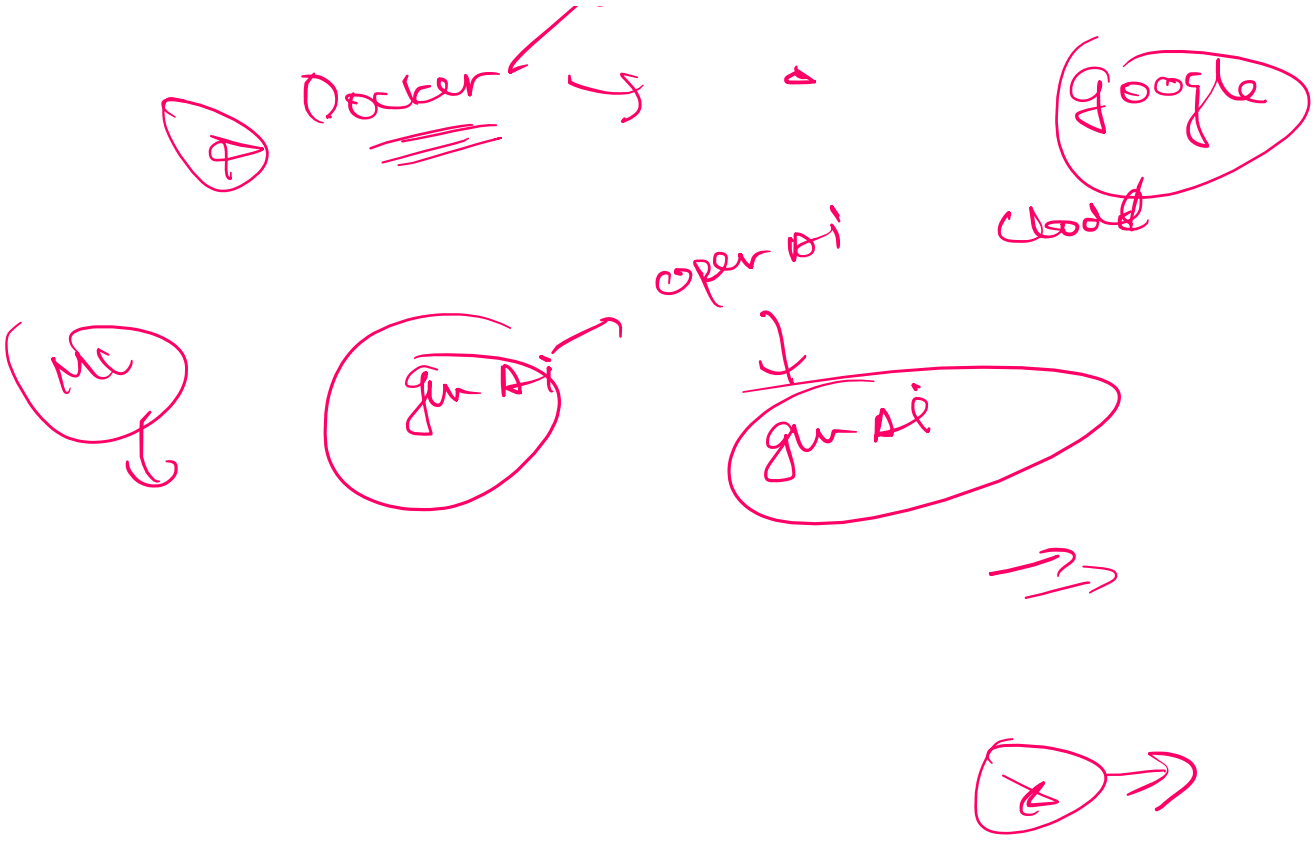
Question

What does the linearity rule of differentiation state?

Choices

- [x] The derivative of a sum of functions is the sum of their derivatives.
- [ ] The derivative of a product of functions is the product of their derivatives.
- [ ] The derivative of a composite function is the composition of their derivatives.
- [ ] The derivative of a quotient of functions is the quotient of their derivatives.

title: Using derivatives for optimization


Docker
Google
Cloud
oper AI
gen AI
gen AI